

Week 15 Capstone — Dataset Selection Guide

Your team picks its own dataset. This is guidance, not a checklist you need approval on — there's no gate to pass before you start. Use it to avoid picking something that turns out not to work halfway through Week 15.

What your data needs to support

Your capstone applies four methods to your data. Before committing to a dataset, check it can support all four:

Method	What you need
Regression (Section B)	A numeric predictor (X) and a numeric outcome (Y) you have a real reason to expect are related
Hypothesis test (Section C)	A categorical or binary grouping variable that splits your rows into two meaningful groups
Monte Carlo simulation (Section D)	Enough rows and enough understanding of your predictor's natural variability to justify an uncertainty assumption
Bootstrap CI (Section B4–B5)	Enough rows for resampling to be meaningful — 100+ rows recommended ; more is better

If your dataset is missing one of these (e.g., no natural way to split into two groups), you can often still make it work by deriving one — a continuous variable can be split at its median, a date column can be split into before/after some event, etc. Think about this before you're deep into the notebook.

Suggested public data sources

These are starting points, not a restricted list — any legitimate public dataset works.

Transportation: - ADOT Open Data Portal — Arizona traffic counts, crash data - FHWA / Bureau of Transportation Statistics — national traffic, freight data - City of Tucson open data portal — local traffic counts, transit ridership

Water resources / hydrology: - USGS National Water Information System — streamflow, groundwater levels - NOAA Climate Data Online — precipita-

tion, weather records (same source as Weeks 1–2’s Tucson data) - Bureau of Reclamation — reservoir levels, water use

Structural / geotechnical: - USGS Earthquake Catalog — seismic event data - State DOT bridge inspection databases (many states publish these) — structural condition ratings over time - National Bridge Inventory — condition, load ratings, age

Energy / buildings: - EIA (Energy Information Administration) — electricity consumption, rates - 5 ready-made building-energy datasets already in the course files (`Week15_Hospital_Dataset.csv` and similar) — no search required

General: - data.gov — federal open data across every domain above
- Kaggle Datasets — searchable, often pre-cleaned (verify the source is legitimate/documented before relying on it)

A few practical notes

- **Download a sample before committing.** Open the CSV, check the actual column names and row count match what you expect — public data portals sometimes have thin documentation.
- **Messy data is normal.** If your dataset needs cleaning (missing values, inconsistent formatting), that’s a legitimate part of real engineering work — just budget time for it.
- **You don’t have to pick something novel.** Using one of the 5 legacy building-energy datasets is a completely valid choice if you’d rather focus your time on the analysis than the search.
- **Ask your team “so what?” before finalizing.** If you can’t articulate why a client would care about the relationship you’re about to model, that’s worth reconsidering before you’re several hours into the notebook.